Green Investment Behaviour: A Firth Logistic Regression Pipeline and the Pitfalls of Replicated Survey Data

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Abstract

This project builds an econometric pipeline to identify factors associated with individual investment in green financial instruments, comparing stated intentions with actual investment. A pilot survey collected 15 responses from Master's students; since this is far too small for regression analysis, the responses were replicated to 97 observations to develop and test the pipeline. Logistic regression suffers from separation, which Firth's bias-reduced estimator addresses, while aggregating rare categories reduces sparsity. Standard cross-validation reports perfect discrimination (AUC = 1), but holding out each distinct response profile entirely shows that this result is driven by data leakage: the AUC falls to 0.67 for stated intention and 0.79 for actual investment, with no meaningful predictive value. The estimated associations are therefore illustrative only, and the project should be read as a methodological exercise on small-sample, rare-event logistic regression and on the risks of replicated data.

1 Introduction

The market for sustainable financial products has grown rapidly, but individual participation remains uneven, and survey evidence suggests a possible gap between what people say they would do and what they actually do. This project was designed to compare the two by modelling two binary outcomes:

- `y.ipo`: the stated intention to invest part of one's capital in green instruments;
- `y.real`: actual investment in green instruments.

The aim was to identify factors associated with these outcomes, which could then motivate future causal research on whether any of them actually affects green investment behaviour. The candidate factors cover four potential barriers: limited capital, low propensity to invest, low environmental sensitivity or information, and distrust in green products.

Given the available data (Section 2), the contribution of the project is methodological: it develops a complete estimation pipeline for small samples with rare events and shows how replicated observations can produce spurious, apparently perfect results.

2 Data

A pilot survey collected **15 responses** from Master's students. Eight respondents stated an intention to invest in green instruments and two reported actually investing in them. The ques-

tionnaire covered capital invested, willingness to invest an unexpected windfall, environmental sensitivity, information on green initiatives, knowledge of the Paris Agreement, perceived impact and safety of green products, and the return respondents would sacrifice for a green instrument (Appendix A).

Since 15 observations are far too few to estimate a regression with 14 regressors, the responses were **replicated to 97 observations**, each response appearing between 3 and 11 times, in order to develop and test the estimation pipeline. The replicated dataset contains only 15 distinct answer profiles, and the 12 "investors" in it correspond to the same 2 real respondents. All statistical results below must be read with this in mind: replication does not add information, but it makes estimates look far more precise than they are.

Preprocessing. All answers were recoded into dummy variables, with one baseline category per question. Perfectly correlated and constant dummies were removed, two adjacent categories of the investment-share question were merged, and remaining exact linear dependencies were identified and removed with `findLinearCombos` in R, leaving 14 regressors.

3 Methodology

Estimation. Both outcomes are modelled with logistic regression. Standard maximum likelihood estimation failed because of separation: some combinations of regressors perfectly predict the outcome, so coefficients diverge. Firth's bias-reduced logistic regression [1] adds a penalty to the likelihood that keeps estimates finite. Since many categories remained very sparse, a second specification merged rare dummies (categories with fewer than 10% of observations in either state).

Interpretation. Associations are summarised by odds ratios (OR) and average marginal effects (AME), the average change in predicted probability, in percentage points, associated with each regressor. Neither should be read causally.

Evaluation. Performance is measured by the AUC (discrimination), the Brier score (calibration) and Tjur's R^2, both in sample and with stratified 5-fold cross-validation. To check whether the replicated rows distort these results, a **leave-one-profile-out cross-validation** was added: each of the 15 distinct response profiles, with all its copies, is held out in turn, the model is estimated on the others, and the AUC is computed on the pooled out-of-sample predictions.

4 Results

4.1 Estimated associations

Table 1 reports the strongest associations from the aggregated specification. Stated intention is associated with general willingness to invest: respondents who would invest 26–50% of an unexpected windfall in financial instruments are much more likely to state they would invest green. For actual investment, very high environmental concern shows the strongest association: in the pilot data, both real investors report it, and no other respondent does.

Outcome	Variable	OR	AME (pp)	p-value
Stated intention (`y.ipo`)	Windfall invested 26–50%	165	+48	< 0.001
Actual investment (`y.real`)	Very high environmental concern	132	+82	0.112

Table 1: Main Firth logistic regression estimates (aggregated specification, replicated data).

These estimates are illustrative only. Each real answer is counted on average about 6.5 times, so standard errors are artificially small and p-values overstate the evidence; with the 15 real responses, and only 2 real investors, none of these patterns can be considered statistically established.

4.2 Model evaluation and data leakage

Table 2 shows why the evaluation needs particular care. In sample and in standard cross-validation, both models reach an AUC of 1 with near-zero Brier scores, which would suggest perfect prediction. Holding out each response profile entirely, the AUC drops to 0.67 for stated intention and 0.79 for actual investment.

	Stated intention (`y.ipo`)		Actual investment (`y.real`)	
	AUC	Brier	AUC	Brier
In sample	1.000	0.005	1.000	0.005
Stratified 5-fold CV	1.000	0.010	1.000	0.010
Leave-one-profile-out CV	0.667	0.275	0.794	0.097
Constant prediction (sample share)	–	0.247	–	0.108

Table 2: Model performance (Firth logistic regression, non-aggregated specification).

The difference has a simple explanation. In standard cross-validation, copies of the same response fall into both the training and the test folds, so the model is evaluated on observations it has already seen: a form of data leakage. Once copies are kept together, the model for stated intention performs worse than a constant prediction equal to the share of positive answers (Brier score 0.275 against 0.247), i.e. it has no predictive value. For actual investment, the Brier score is only marginally better than the constant benchmark, and the result rests on just two distinct investor profiles.

5 Discussion and Limitations

The project delivers a complete pipeline for binary survey outcomes in small samples with rare events: preprocessing, detection of separation, Firth estimation, aggregation of sparse categories, interpretation through odds ratios and marginal effects, and evaluation in and out of sample. Its most important lesson, however, concerns the data. Replicating observations does not increase information: it produces separation, artificially small standard errors and perfect cross-validated scores that disappear as soon as the evaluation accounts for the duplicated profiles.

The substantive question therefore remains open. The pilot responses hint that stated intentions may reflect general investment readiness, while actual investment may be more closely related to strong environmental values, but with 15 respondents these are anecdotal observations. Testing them would require a substantially larger survey of genuine responses; only then could the associations identified here be used to design a causal study of the drivers of green investment.

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References

[1] D. Firth. Bias reduction of maximum likelihood estimates. *Biometrika*, 80(1):27–38, 1993.

A Variable Coding

Survey question	Response categories	Variable prefix
Q1: Share of own capital invested in financial instruments	0% (baseline), 1–25%, 26–50%, 51–75%, >75%	`perc.inv`
Q2: Share of an unexpected windfall (e.g. a gift) willing to invest	0% (baseline), 1–25%, 26–50%, 51–75%, >75%	`perc.reg`
Q3: Environmental sensitivity	None (baseline), low, medium, high	`env`
Q4: Information on green initiatives	None (baseline), low, medium, high	`info`
Q5: Knowledge of the Paris Agreement	None (baseline), low, medium, high	`paris`
Q6: Belief that green investments have real impact	None (baseline), low, medium, high	`imp`
Q7: Perceived safety of green vs. traditional instruments	None (baseline), low, medium, high	`safe`
Q8: Maximum annual return willing to sacrifice for a green instrument	0% (baseline), 0–1%, 1–2%, 2–2.5%, >2.5%	`max`
Q9: Would you invest part of your capital in green instruments?	Yes / No	`y.ipo`
Q10: Do you already invest in green instruments?	Yes / No	`y.real`

Table 3: Coding of survey responses into dummy variables (suffixes `.little`, `.some`, `.very` denote low, medium and high).